Virtual environments

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Why do we need virtual environments?



Software dependency hell

https://en.wikipedia.org/wiki/Dependency_hell



Why do we need virtual environments?

- Avoiding software dependency hell
- Isolation of external code from the PC
- Easy to install/move/remove



Which virtual environments do we have?

- venv
- virtualenv + virtualenvwrapper
- conda / mamba
- renv



Which virtual environments do we have?

- **venv**
- virtualenv + virtualenvwrapper
- conda / mamba
- renv



Creating virtual environment

Hands on your keyboard

```
python3 -m venv my_venv
```

```
ls -al
```

```
ls -al my_venv
```

What do you see?



Using virtual environment

Hands on your keyboard

We can activate it and deactivate

```
which python
```

```
source my_venv/bin/activate
```

```
which python
```

```
deactivate
```

```
which python
```



Using virtual environment

Hands on your keyboard

Let’s install some library!

```
source my_venv/bin/activate
```

```
ls my_venv/lib/python3.11/site-packages
```

```
pip install cowsay
```

```
my_venv/bin/cowsay -t hello
```



Using virtual environment

Hands on your keyboard

Let’s install some library!

```
pip install biopython
```

numpy installed as well!

```
python3
```

```
from Bio.Seq import Seq
my_dna = Seq('AGTACACTGGT')
print(my_dna.complement())
quit() / Ctrl + D
```



Sharing virtual environment

Hands on your keyboard

```
pip freeze
```

Saving info about our venv

```
deactivate
```

```
pip freeze
```

```
source my_venv/bin/activate
```

```
pip freeze …
```



Sharing virtual environment

Hands on your keyboard

```
pip freeze
```

Saving info about our venv

```
deactivate
```

```
pip freeze
```

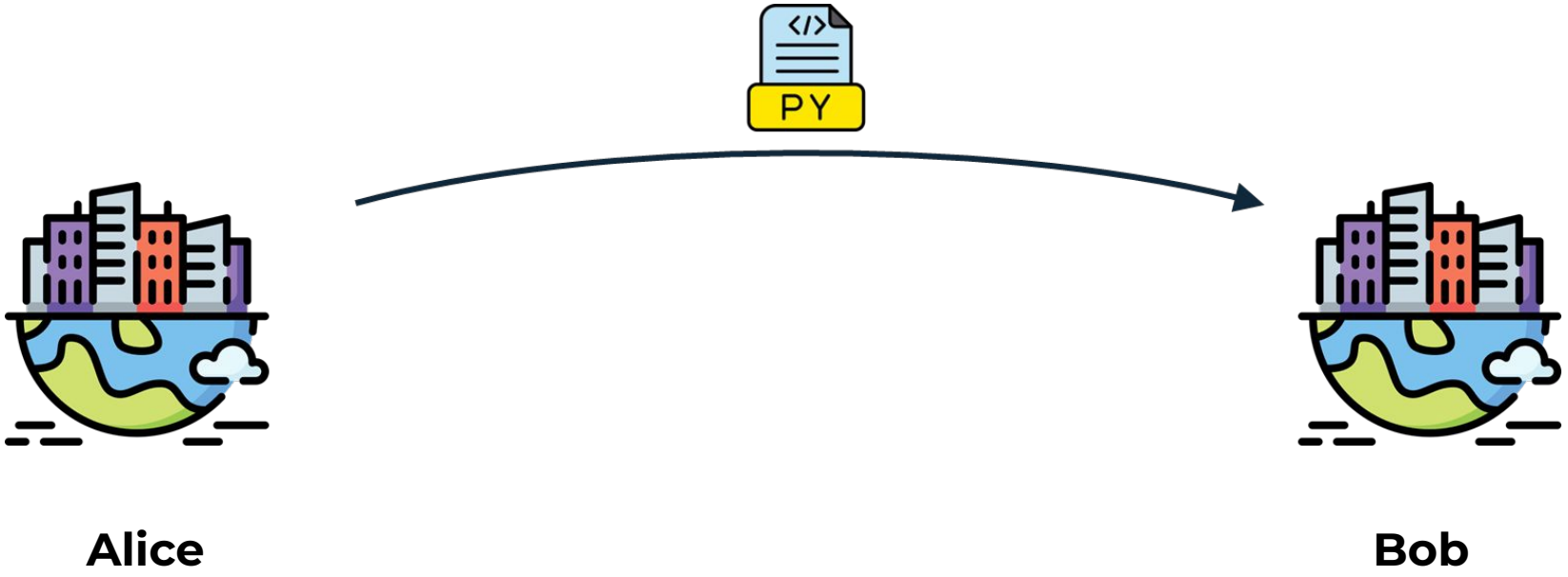
```
source my_venv/bin/activate
```

```
pip freeze > requirements.txt
```

```
cat requirements.txt
```



Sharing virtual environment



Sharing virtual environment

Hands on your keyboard

Reproducing venv

```
python3 -m venv my_venv2
```

```
source my_venv2/bin/activate
```

```
ls my_venv2/lib/python3.11/site-packages
```

```
pip install -r requirements.txt
```

```
ls my_venv2/lib/python3.11/site-packages
```



Deleting virtual environment



Deleting virtual environment

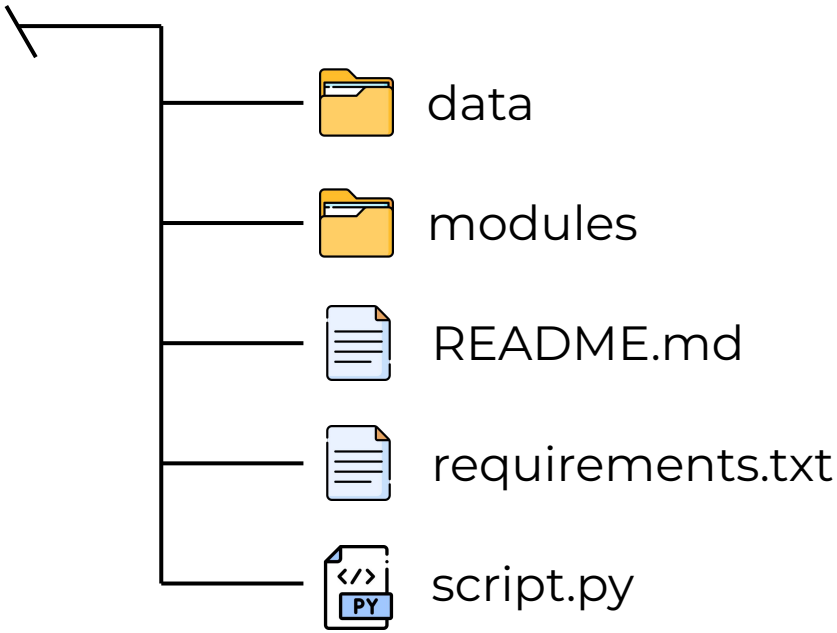
Hands on your keyboard

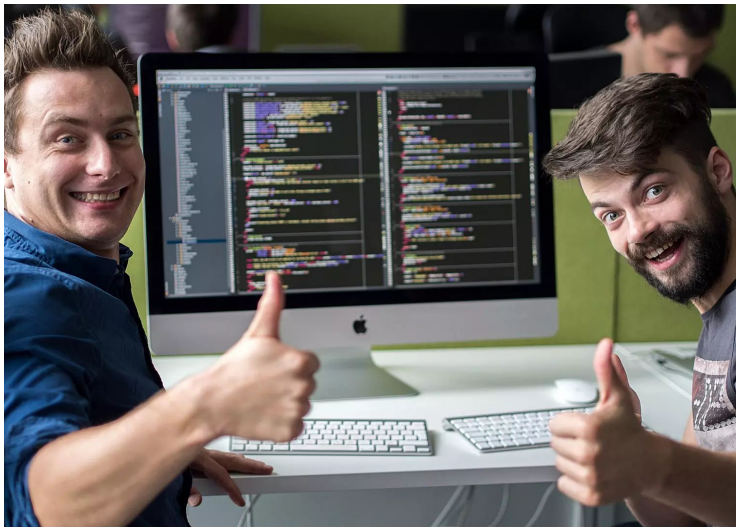
Don’t forget to

```
deactivate
```

```
rm -r my_venv2
```

Example bioinformatics project





Which virtual environments do we have?

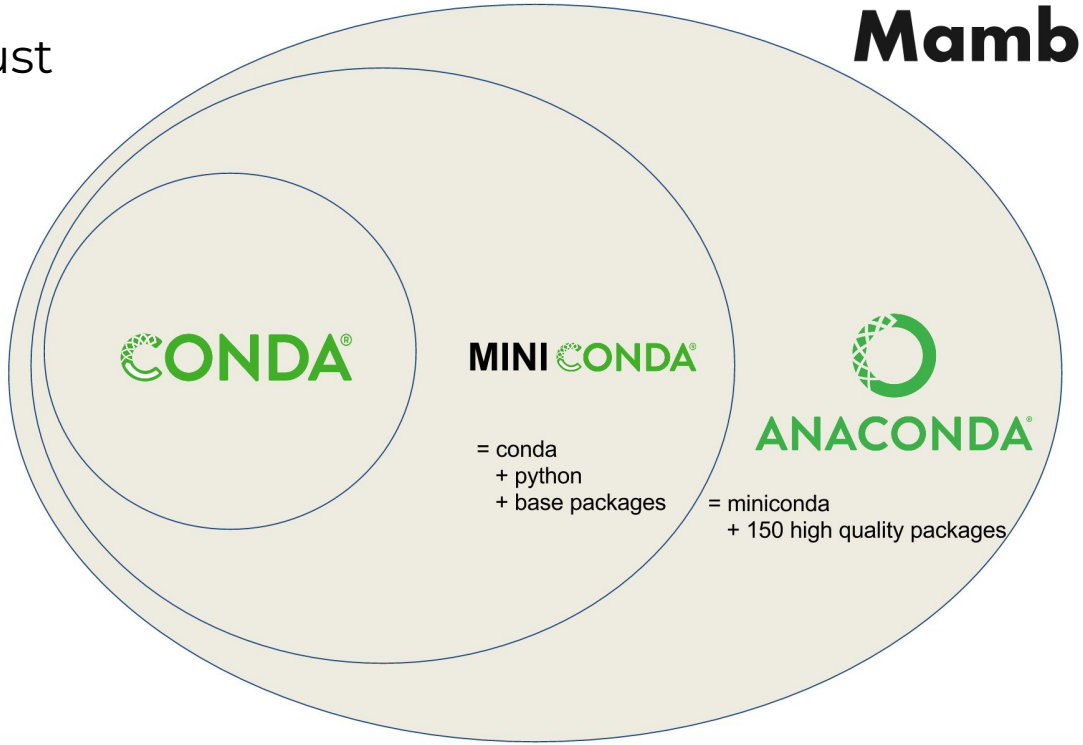
- venv
- virtualenv + virtualenvwrapper
- **conda / mamba**
- renv



Conda environments

To manage more than just python packages

Mamba



Two conda strategies

When to use environments? → **Always!**

When to create new environments?

- For every tool
- For every project





Using conda/mamba

Hands on your keyboard

Creating environments

```
conda create -n new_env
```

```
conda activate new_env
```

```
conda deactivate
```

People often make aliases *ca* and *cda*



Using conda/mamba

Hands on your keyboard

Inspecting environments

```
conda env list
```

```
conda list
```

```
conda info
```

```
conda env export
```

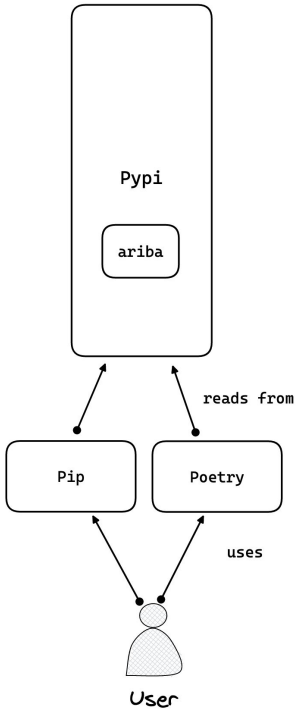


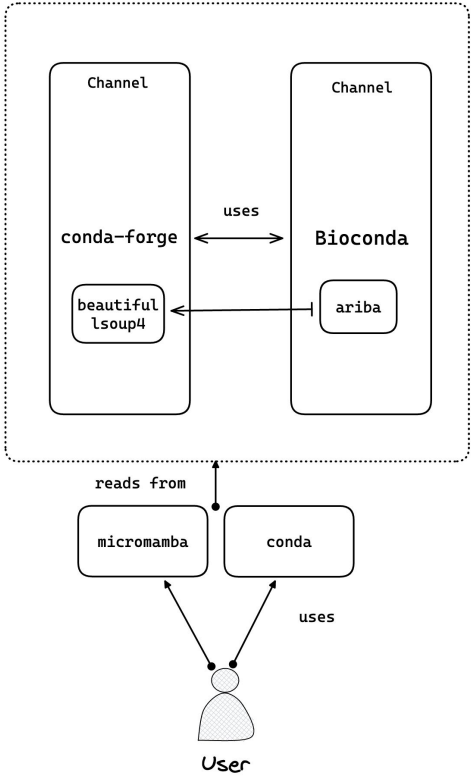
Conda channels

Conda channels are remote repositories with packages

CONDA-FORGE

BIOCONDA®





Using conda/mamba

Hands on your keyboard

Sharing environments

```
conda install -c channel
```

```
conda env export > environment.yaml
```

```
conda env create -f environment.yaml
```

```
conda env export --from-history
```



Wrap up

Make and use virtual environments

Always make requirements or yaml

